

Introduction to Virtual Reality, 2026 Spring

Assignment 3 / Due end of semester

Google Cardboard HMD: <https://www.youtube.com/watch?v=EHkOnsvpHiA>

1. Use two magnifying lenses (purchasable at a reasonably cheap price from different places such as <http://www.iloupe.kr/>). I can reimburse you for any cost!
2. Draw out the optical diagram and viewing model and a blue print for your design.
 - A. Determine size and power of magnifying lens.
 - B. Determine the distance from eye to lens and lens to display.
 - C. Estimate the perceived distance to the magnified imagery.
 - D. Render a stereoscopic picture based on the viewing model (generate two images for left and right to be displayed on your phone).
3. Make a short report. Include observations on:
 - A. Image distortion (e.g. barrel distortion?)
 - B. Field of view (from different relief distances).
 - C. Sickness or fatigue from lens distortion.
 - D. Image resolution or quality and one's visual acuity
 - E. Any others.